

Exercise Sheet 4

Straight Line Complexity of Linear Programs

Algorithms and Geometry of Linear Programming — DIMAG Summer School, August 2026

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This sheet restates the exercises of Lecture 4. Throughout, we work with the standard-form pair

$$\min c^\top x \text{ s.t. } Ax = b, x \geq 0, \quad \max b^\top y \text{ s.t. } A^\top y + s = c, s \geq 0,$$

with $A \in \mathbb{R}^{m \times n}$ of full row rank, both programs strictly feasible, and common optimal value v^* ; the gap formula reads $c^\top x - b^\top y = s^\top x \geq 0$. The *max central path* is $x_i^m(g) := \max\{x_i : x \text{ feasible, } c^\top x \leq v^* + g\}$ — nondecreasing, concave, piecewise linear — and the *straight line complexity* $\text{SLC}_\gamma(f)$ of a nonnegative f is the least number of pieces of a piecewise linear h with $\gamma f \leq h \leq f$. A *circuit* of A is a support-minimal nonzero kernel vector.

Exercise 1 (The max central path of maximum flow). Let $\mathcal{P} \subseteq \mathbb{R}_+^E$ be the feasible region of the maximum flow problem of Lecture 3 on a network $G = (V, E)$ with source s , sink t and capacities u , with flow value $\text{val}(f)$ and optimal value v^* . For $e \in E$, the max central path on e is

$$f_e^m(g) := \max\{f_e : f \in \mathcal{P}, \text{val}(f) \geq v^* - g\}.$$

Prove that f_e^m is piecewise linear with at most 2 pieces, i.e., $\text{SLC}_1(f_e^m) \leq 2$. (Hint: Adjacent vertices f^1, f^2 of $\mathcal{P}$ with $\text{val}(f^1) > \text{val}(f^2)$ differ by a t - s path in the residual network $N[f^1]$ of Lecture 3.)

Exercise 2 (Greedy and staircase approximation). Let $f : [0, T] \rightarrow \mathbb{R}_+$ be concave and nondecreasing and let $\gamma \in (0, 1]$. For $0 \leq a < b \leq T$, let $\ell_{a,b}^f$ denote the linear interpolation of f on $[a, b]$, and set $s_b^f(g) := f(b) \min\{g/b, 1\}$ and $s_0^f := f(0)$.

1. Prove that the following greedy construction is optimal, achieving $\text{SLC}_\gamma(f, T)$ pieces: build breakpoints $0 = g_0 < g_1 < \dots < g_k = T$, taking $g_i \in (g_{i-1}, T]$ maximal with $\gamma f \leq \ell_{g_{i-1}, g_i}^f \leq f$ on $[g_{i-1}, g_i]$, and let $h := \ell_{g_{i-1}, g_i}^f$ there. (Hint: For any valid p with breakpoints $0 = b_0 < \dots < b_r = T$, show by induction that $i \leq r$ and $b_i \leq g_i$.)
2. Let $h^s := \max_{0 \leq i \leq k} s_{g_i}^f$. Prove that $\frac{1}{2} h \leq h^s \leq h$ on $[0, T]$: up to a factor 2 in the number of pieces and in the approximation factor, $\text{SLC}_\gamma(f, T)$ is achieved by a staircase.

Exercise 3 (Duality of the max central path). Define the dual max central path by $s_i^m(g) := \max\{s_i : (s, y) \text{ dual feasible, } b^\top y \geq v^* - g\}$.

1. Prove that $g \leq x_i^m(g) s_i^m(g) \leq 2g$ for all $g \geq 0$ — a combinatorial analogue of the centrality equation $x_i s_i = \mu$ of Lecture 5. (Hint: Upper bound: the gap formula. Lower bound: dualize both maximization problems and multiply the optimal values.)
2. Combine part 1 with the staircases of Exercise 2 to convince yourself — no formal proof needed — that x_i^m and s_i^m have the same straight line complexity up to constant factors in both parameters. (Hint: What does $g \mapsto g/s_b^f(g)$ look like?)

Setup: generalized flow. Let $G = (V, E)$ be a directed graph whose edges $e = (i, j)$ carry gain factors $\gamma_e > 0$. The *generalized incidence matrix* $\mathbf{A} \in \mathbb{R}^{V \times E}$ has, in the column of $e = (i, j)$, the entry -1 in row i and γ_e in row j (for $\gamma \equiv \mathbf{1}$, the node-arc incidence matrix of Lecture 3); a kernel vector is a *generalized circulation* — f_e units entering e deliver $\gamma_e f_e$ at its head. The *gain* of a cycle C with a chosen traversal direction is $\gamma(C) := \prod_{e \text{ forward}} \gamma_e \cdot \prod_{e \text{ backward}} \gamma_e^{-1}$, and C is *conservative* if $\gamma(C) = 1$. Finally, a *bicycle* is a connected subgraph with exactly two more edges than a spanning tree: two cycles meeting at one node, two disjoint cycles joined by a path, or a theta graph. When its cycles are non-conservative, a bicycle carries a natural flow: a *flow-generating* cycle ships its surplus along the path to a *flow-absorbing* cycle.

Exercise 4 (Circuits of generalized flow). Show that the supports of the circuits of $\mathbf{A}$ are exactly (i) the conservative cycles, and (ii) the bicycles containing no conservative cycle. (Hint: (i) Push one unit around a cycle C : one lap multiplies the flow by $\gamma(C)$. (ii) Forests, and connected subgraphs whose only cycle is non-conservative, have independent columns. (iii) On a bicycle, generate a surplus on one cycle and absorb it on the other.)